LAB 6: Dynamic Routing Configuration – RIP, EIGRP, OSPF, and BGP

OBJECTIVES:

1. To understand the working principles of dynamic routing protocols including RIP, EIGRP, OSPF, and BGP.
2. To configure dynamic routing protocols on routers using Cisco Packet Tracer.
3. To observe automatic route discovery and routing table updates.
4. To verify communication between different networks using ping testing.

HARDWARE AND SOFTWARE REQUIREMENTS:

- Cisco Packet Tracer (Version 6.2 or higher)
- Windows PC / Laptop

THEORY:

Dynamic routing protocols allow routers to automatically exchange routing information and update routing tables without manual configuration of individual routes. Unlike static routing, dynamic routing adapts to network topology changes.

1. RIP (Routing Information Protocol)

RIP is a **distance-vector routing protocol** that uses hop count as its metric to determine the best path.

Operation:
RIP sends the entire routing table to its directly connected neighbors every 30 seconds. The maximum hop count allowed is 15; a hop count of 16 is considered unreachable.

Advantages:

- Simple to configure
- Suitable for small networks

Limitations:

- Slow convergence
- Limited scalability

2. EIGRP (Enhanced Interior Gateway Routing Protocol)

EIGRP is an advanced distance-vector protocol developed by Cisco. It is often referred to as a hybrid protocol because it combines features of both distance-vector and link-state routing.

Operation:
EIGRP uses the Diffusing Update Algorithm (DUAL) to calculate the shortest path. It sends partial updates only when changes occur, rather than periodic full-table updates.

Advantages:

- Fast convergence
- Efficient bandwidth utilization
- Supports multiple metrics

3. OSPF (Open Shortest Path First)

OSPF is a **link-state routing protocol** that builds a complete map of the network topology.

Operation:
OSPF uses the Dijkstra Shortest Path First (SPF) algorithm to calculate the best route. Routers exchange Link State Advertisements (LSAs) to build a topology database.

Advantages:

- Scalable for large networks
- Supports hierarchical design using Areas
- Faster convergence than RIP

4. BGP (Border Gateway Protocol)

BGP is a **path-vector protocol** used to exchange routing information between different Autonomous Systems (AS).

Operation:
BGP makes routing decisions based on path attributes, policies, and network rules rather than simple technical metrics like hop count.

Application:
BGP is the primary routing protocol used across the Internet (ISP-to-ISP communication).

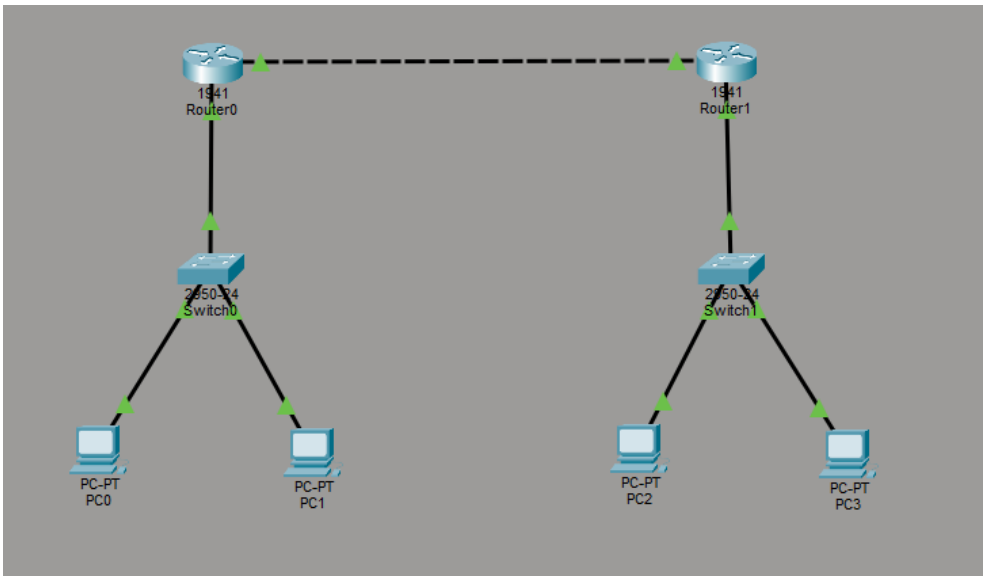
NETWORK TOPOLOGY:

Both routers were connected through a serial/WAN link using the network **10.0.0.0**.

Two separate LAN networks were created:

- **Network 1:** 192.168.1.0/24 connected to Router0 via Switch0 (PC0 and PC1)
- **Network 2:** 192.168.2.0/24 connected to Router1 via Switch1 (PC2 and PC3)

Each LAN had its own default gateway configured on the router interface.



IP CONFIGURATION SUMMARY:

Device	Interface	IP Address	Subnet Mask
Router0	G0/1	192.168.1.1	255.255.255.0
Router0	G0/0	10.0.0.1	255.0.0.0
Router1	G0/1	192.168.2.1	255.255.255.0
Router1	G0/0	10.0.0.2	255.0.0.0
PC0	FastEthernet0	192.168.1.2	255.255.255.0
PC1	FastEthernet0	192.168.1.3	255.255.255.0
PC2	FastEthernet0	192.168.2.2	255.255.255.0
PC3	FastEthernet0	192.168.2.3	255.255.255.0

IMPLEMENTATION OF DYNAMIC ROUTING PROTOCOLS

1. RIP Configuration Commands

Router0:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

Router1:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

2. EIGRP Configuration Commands

Router0:

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

Router1:

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.2.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

3. OSPF Configuration Commands

Router0:

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

Router1:

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

4. BGP Configuration Commands

Router0 (AS 65001):

```
Router(config)# router bgp 65001
Router(config-router)# neighbor 10.0.0.2 remote-as 65002
Router(config-router)# network 192.168.1.0 mask 255.255.255.0
```

Router1 (AS 65002):

```
Router(config)# router bgp 65002
Router(config-router)# neighbor 10.0.0.1 remote-as 65001
Router(config-router)# network 192.168.2.0 mask 255.255.255.0
```

RESULT:

After configuring each protocol individually, connectivity was tested using ping commands.

1. RIP Configuration

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

2. EIGRP Configuration

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3. OSPF Configuration

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

4. BGP Configuration

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

Routing tables were updated automatically in each case, and packets were transmitted successfully with no loss (except some minor timeout behavior observed during initial establishment)

This confirms that all dynamic routing protocols were functioning correctly.

DISCUSSION & CONCLUSION:

During this lab session, the concept of dynamic routing was implemented to understand how routers automatically exchange routing information and update their routing tables without manual intervention. By configuring RIP, EIGRP, OSPF, and BGP individually, we were able to observe the differences in routing behavior, convergence speed, and protocol functionality. RIP demonstrated basic distance-vector routing with periodic updates and hop-count limitations. EIGRP showed faster convergence and efficient bandwidth usage due to the Diffusing Update Algorithm (DUAL). OSPF provided a clear understanding of link-state routing, where routers build a topology database and calculate the shortest path using the SPF algorithm. BGP introduced the concept of inter-autonomous system routing and policy-based path selection, which is widely used across the Internet.

The successful ping tests between PC0 (192.168.1.2) and PC2 (192.168.2.2) confirmed that all routing protocols were functioning correctly and that routing tables were updated dynamically. Hence, the lab was completed successfully with proper understanding and implementation of dynamic routing protocols using Cisco Packet Tracer.